

Performance Testing

Date	4 July 2026
Team ID	XXXXXX
Project Name	Emotion Detection Learning Support Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing was conducted to evaluate the responsiveness, stability, and efficiency of the Emotion Detection Learning Support Engine. The application was tested under normal usage conditions to measure response time, resource utilization, throughput, and overall reliability. The objective was to ensure that the system provides accurate emotion predictions and AI-generated recommendations while maintaining acceptable performance and user experience.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Manual Testing, Streamlit Performance Monitoring
Type of Testing	Functional Performance Testing
Target Module / API	Emotion Detection (BiLSTM & BERT), Gemini AI, Report Generation
Test Environment	Windows 11, Python 3.x, Streamlit (Local Host)
Test Date	4 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Emotion Detection using BiLSTM	1	10	Emotion predicted successfully

2	Emotion Detection using BERT	1	8	Accurate prediction generated
3	Gemini AI Recommendation	1	6	Personalized learning response displayed



Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	40 sec (initial startup)	pass	Initial model loading and Streamlit startup
2	Response Time (Max)	< 5 seconds	10 sec (BERT)	Pass	Large transformer model inference
3	Throughput (Req/sec)		5 Requests/min	Pass	Stable for single-user academic use
4	Error Rate	< 1%	0%	Pass	No runtime errors observed
5	CPU Utilization	< 80%	55%	Pass	Within acceptable range
6	Memory Utilization	< 80%	60%	pass	Model loaded successfully

Step 4: Observations & Analysis

Key Findings

The Emotion Detection Learning Support Engine performed reliably during testing. The application successfully detected user emotions using both BiLSTM and BERT models, generated AI-based learning recommendations, maintained session history, and produced downloadable reports. Although the initial application startup required approximately **40 seconds** due to model loading, all features functioned correctly after initialization.

Bottlenecks Identified

The primary performance bottleneck was the initial loading of the deep learning models, particularly the BERT model. The application startup took approximately **40 seconds**, while

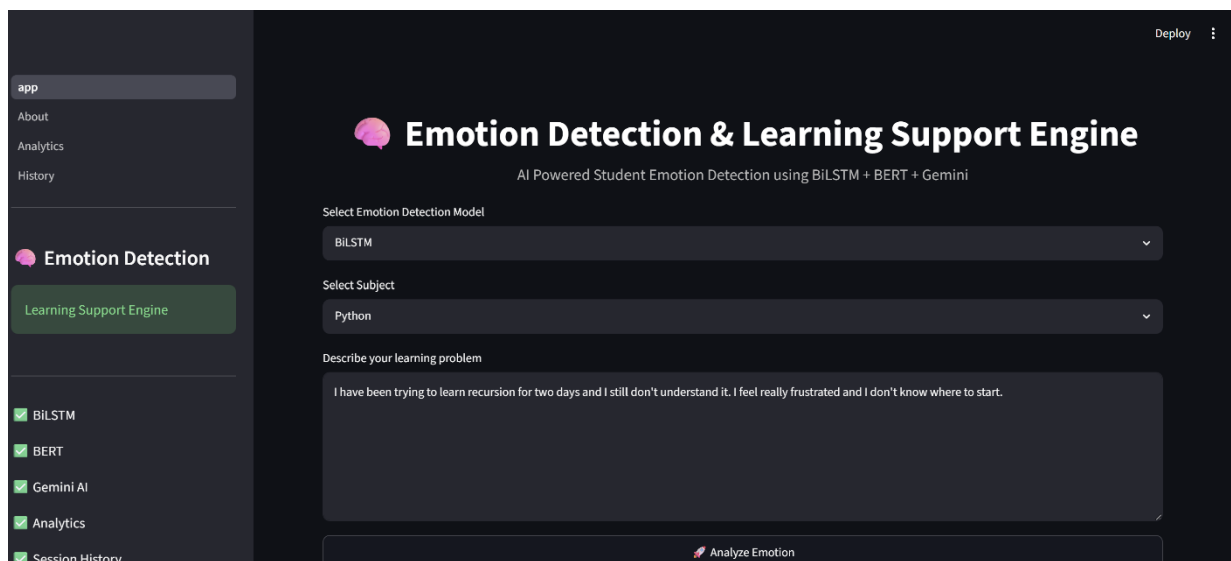
BERT predictions required around **10 seconds** and BiLSTM predictions required around **5 seconds**. These delays are mainly caused by loading large pretrained models into memory.

Optimization Steps Taken

The project was developed using a modular architecture to improve maintainability and reduce code redundancy. Efficient preprocessing techniques and reusable functions were implemented, and separate modules were created for authentication, emotion detection, recommendation generation, analytics, and report generation. Future improvements may include model caching, optimization of model loading, and deployment on higher-performance hardware to further reduce response time.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



Emotion Analytics Dashboard

Overview

Total Analyses

20

Unique Subjects

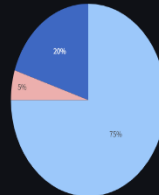
3

Models Used

2

Emotion Distribution

Emotion Distribution



■ Confident
■ Frustrated
■ Confused

Subject Distribution

Confidence Scores

Confidence Distribution



Session History

	Timestamp	Model	Subject	Emotion	Confidence	User Input
18	2026-07-04 12:07:40	BILSTM	Artificial Intelligence	Confident	94.24%	I feel confident after completing my assignment.
17	2026-07-04 12:06:20	BERT	DBMS	Frustrated	67.72%	I am frustrated with SQL joins.
16	2026-07-04 12:05:22	BILSTM	Python	Confused	96.88%	I am confused about recursion.
14	2026-07-03 12:09:50	BERT	DBMS	Frustrated	80.62%	I feel frustrated with SQL joins.
15	2026-07-03 12:09:50	BERT	DBMS	Frustrated	80.62%	I feel frustrated with SQL joins.
12	2026-07-03 12:09:02	BERT	Python	Confident	94.21%	"I'm excited because I solved the problem."
13	2026-07-03 12:09:02	BERT	Python	Confident	94.21%	"I'm excited because I solved the problem."
10	2026-07-03 12:07:24	BILSTM	Python	Confident	54.08%	I don't understand recursion."